

LEIS COMPLIANCE & TRUSTWORTHINESS MATRIX

Formal Mapping of LEIS Architectural Safeguards to the European Union AI Act and W3C PROV-O Standards

1. Regulatory Alignment Abstract

Under the strict compliance mandates of the European Union (EU) AI Act and national Czech regulatory directives, any artificial intelligence system deployed in high-risk enterprise, administrative, or critical public sectors must satisfy absolute benchmarks of **traceability, explainability, human oversight, and data provenance**. Traditional generative models operate as closed-source black boxes, outputting unvalidated probabilistic text. LEIS is engineered from the ground up to solve this compliance crisis. It acts as an open, auditable metadata layer that enforces mathematical alignment, logs decision lineages, and guarantees that every machine action is traceably linked back to human intent and verifiable physical reality.

2. EU AI Act High-Risk Requirements Mapping

Act Article	Requirement Domain	Traditional LLM Failure	LEIS Compliance Mechanism
Article 9	Risk Management System	Silent hallucination propagation across system calls.	Conflict Capsules & Drift Engine: Instantly quarantines divergent states, halting model output.
Article 12	Record-Keeping & Logging	Static token-log dumps containing no structural reasoning.	W3C PROV Metadata: Automated generation of unalterable, structured action provenance.
Article 13	Transparency & Information	Over-confident, opaque answers with zero confidence metrics.	The Humanizer Layer: Decouples raw data from understanding, displaying gaps and limits.
Article 14	Human Oversight	Autonomous loops that cannot be intercepted by human supervisors.	RCT & Socratic Prompts: Mandates validation gates, returning control to humans during drift.
Article 15	Accuracy & Robustness	Susceptible to input manipulation and adversarial injection.	Immutable Root Seeds: Validates predictions against physical, verified Reality Markers.

3. The W3C PROV-O Standard Compatibility Layer

To ensure seamless integration with international semantic web registries, LEIS implements an un-aliased compatibility layer mapping its native core concepts to the W3C PROV Ontology. This guarantees that LEIS files are readable by any enterprise ontology parser on Earth.

LEIS Concept	W3C PROV-O Equivalent	Semantic Definition & Boundary
Reality Source	prov:Entity / prov:Agent	The baseline physical origin or creator supplying raw validation telemetry.
Artifact	prov:Entity	A physical container or document manifested from reality (e.g. PDF, CSV, Registry).
Reality Object	prov:Entity (Extended)	The stable, 4D identity of an observed reality independent of its file format.
Capsule	prov:Entity (Composite)	The structured, compressed packet representing a single resolved state of understanding.
Lineage Event	prov:Activity	The unalterable event tracing how an understanding evolved or transitioned.
AI System	prov:SoftwareAgent	The machine prediction engine acting as an execution and reconstruction collaborator.